Solving a 2D EM simulation with PINN

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1 Introduction

Electromagnetic wave behavior in complex materials is described by Maxwell's equations. Under time-harmonic conditions, when wave oscillations are steady over time, these equations reduce to the Helmholtz equation.

In this project, the objective is to solve the two-dimensional Helmholtz equation in the presence of a localized Gaussian source with standard deviation $\sigma = 0.05$ and spatially varying permittivity.

Rather than relying on traditional numerical techniques such as finite-difference or finite-element methods, which require spatial discretization and can be computationally demanding, we adopt a Physics-Informed Neural Network (PINN) approach. PINNs provide a mesh-free, data-efficient alternative by training a neural network to approximate a solution that satisfies the governing physics. This is achieved by enforcing the PDE constraints at selected collocation points using automatic differentiation and gradient-based optimization.

The main objective of this project is to solve the two-dimensional Helmholtz equation, at a normalized angular frequency $\omega = 12.0$, arising from electromagnetic wave propagation in complex media using a Physics-Informed Neural Network (PINN). The approach begins with deriving the governing equation from Maxwell's equations. Following this, a PINN is implemented using the SIREN (Sinusoidal Representation Networks) architecture. Finally, the resulting solution is analyzed to verify its physical consistency.

2 Deriving the Helmholtz Equation from Maxwell's Equations

The derivation begins with Maxwell's equations in a source-driven, linear, isotropic, and non-magnetic medium:

Gauss's Law

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon} \tag{1}$$

Gauss's Law for Magnetism

$$\nabla \cdot \mathbf{B} = 0 \tag{2}$$

Faraday's Law

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t} \tag{3}$$

Ampère–Maxwell Law

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} + \mathbf{J} \tag{4}$$

By taking the curl of Faraday's law (3), we get:

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{E}) &= \nabla \times \left(-\mu_0 \frac{\partial \mathbf{H}}{\partial t} \right) \Rightarrow \\ \nabla \times (\nabla \times \mathbf{E}) &= -\mu_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) \end{aligned} \tag{5}$$

Substituting Ampère's Law (4) into the equation above (5), we obtain:

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{E}) &= -\mu_0 \frac{\partial}{\partial t} \left(\epsilon \frac{\partial \mathbf{E}}{\partial t} + \mathbf{J} \right) \Rightarrow \\ \nabla \times (\nabla \times \mathbf{E}) &= -\mu_0 \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} - \mu_0 \frac{\partial \mathbf{J}}{\partial t} \end{aligned} \tag{6}$$

Applying the vector identity

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \tag{7}$$

For a charge-free region, we have $\nabla \cdot \mathbf{E} = 0$, so the identity simplifies to:

$$\begin{aligned} \nabla \times (\nabla \times \mathbf{E}) &= \nabla(\cancel{\nabla \cdot \mathbf{E}}) - \nabla^2 \mathbf{E} \\ \nabla \times (\nabla \times \mathbf{E}) &= -\nabla^2 \mathbf{E} \end{aligned} \tag{8}$$

Substituting (8) into (6), the time-domain wave equation is obtained:

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial \mathbf{J}}{\partial t} \tag{9}$$

Assume a time-harmonic electric field expressed as:

$$\mathbf{E}(\mathbf{r}, t) = \Re \left\{ \mathbf{E}(\mathbf{r}) e^{-i\omega t} \right\} \tag{10}$$

where $\mathbf{E}(\mathbf{r})$ is the complex spatial phasor. This field is real-valued and represents the measurable physical quantity.

We now compute the first and second time derivatives of $\mathbf{E}(\mathbf{r}, t)$:

$$\frac{\partial \mathbf{E}(\mathbf{r}, t)}{\partial t} = \Re \left\{ \frac{\partial}{\partial t} \left[\mathbf{E}(\mathbf{r}) e^{-i\omega t} \right] \right\} = \Re \left\{ -i\omega \mathbf{E}(\mathbf{r}) e^{-i\omega t} \right\} \tag{11}$$

$$\frac{\partial^2 \mathbf{E}(\mathbf{r}, t)}{\partial t^2} = \Re \left\{ \frac{\partial^2}{\partial t^2} \left[\mathbf{E}(\mathbf{r}) e^{-i\omega t} \right] \right\} = \Re \left\{ (-i\omega)^2 \mathbf{E}(\mathbf{r}) e^{-i\omega t} \right\} = \Re \left\{ -\omega^2 \mathbf{E}(\mathbf{r}) e^{-i\omega t} \right\} \tag{12}$$

Both time derivatives of the real field correspond to taking derivatives of the complex exponential, and then taking the real part of the result.

To simplify calculations, we work directly with the complex field $\mathbf{E}(\mathbf{r})e^{-i\omega t}$ and apply differential operators to it. This is valid due to the linearity of Maxwell's equations and differential operators. After solving the equations in the frequency domain, the physical, real-valued field is recovered by taking the real part:

$$\mathbf{E}_{\text{real}}(\mathbf{r}, t) = \Re\left\{\mathbf{E}(\mathbf{r})e^{-i\omega t}\right\} \tag{13}$$

This approach encodes both amplitude and phase information and allows the time derivatives to be replaced algebraically by:

$$\frac{\partial}{\partial t} \longrightarrow -i\omega, \qquad \frac{\partial^2}{\partial t^2} \longrightarrow -\omega^2 \tag{14}$$

within the complex domain.

Therefore equation (9) takes the following form

$$\nabla^2 \mathbf{E} + \omega^2 \mu_0 \epsilon \mathbf{E} = -i\omega\mu_0 \mathbf{J} \tag{15}$$

In this work, we adopt normalized units with $\mu_0 = 1$. Under this choice, the frequency-domain Helmholtz equation becomes:

$$\nabla^2 \mathbf{E} + \omega^2 \epsilon \mathbf{E} = -i\omega \mathbf{J} \tag{16}$$

This is the form directly implemented in the Physics-Informed Neural Network solver.

3 Transverse Electric (TE) Polarization and 2D Reduction

The vector wave equation derived in the previous section (15) can be further simplified under the assumption of *Transverse Electric (TE)* polarization in a two-dimensional setting. In this configuration, the fields are assumed to be invariant along the z-axis:

$$\frac{\partial}{\partial z} = 0 \tag{17}$$

This assumption is appropriate for planar structures such as waveguides or metasurfaces, where wave propagation occurs in the x–y plane and the structure is uniform in z.

In TE polarization, by definition, the electric field has no components in the direction of propagation (i.e., in-plane), and only the out-of-plane component E_z is non-zero:

$$\mathbf{E}(x, y) = E_z(x, y)\,\hat{z}, \qquad \mathbf{H}(x, y) = H_x(x, y)\,\hat{x} + H_y(x, y)\,\hat{y} \tag{18}$$

Similarly, the impressed source current is assumed to be polarized in the same direction:

$$\mathbf{J}(x, y) = J_z(x, y)\,\hat{z} \tag{19}$$

Substituting equations (18) and (19) into the vector Helmholtz equation:

$$\nabla^2 \mathbf{E} + \omega^2 \mu_0 \epsilon(x, y) \, \mathbf{E} = -i\omega\mu_0 \mathbf{J} \tag{20}$$

and recognizing that only the z-components are non-zero, we obtain the scalar Helmholtz equation for E_z:

$$\nabla^2 E_z(x, y) + \omega^2 \mu_0 \epsilon(x, y) \, E_z(x, y) = -i\omega\mu_0 J_z(x, y) \tag{21}$$

This equation governs the spatial behavior of the out-of-plane electric field E_z in the presence of a source term J_z and spatially varying relative permittivity $\epsilon(x, y)$.

Finally, under a normalized unit system where the magnetic permeability is set to $\mu_0 = 1$, the equation simplifies to:

$$\boxed{\nabla^2 E_z(x, y) + \omega^2 \epsilon(x, y) \, E_z(x, y) = -i\omega J_z(x, y)} \tag{22}$$

We impose an open boundary via the Sommerfeld radiation condition:

$$\left(\frac{\partial}{\partial n} + ik \right) E_z = 0, \tag{23}$$

applied on the square boundary of the domain, with $k = \omega\sqrt{\epsilon}$. In our setup the boundary lies in free space, so $\epsilon = 1$ and $k = \omega$.

Equation (22) is the fundamental PDE that we aim to solve using a Physics-Informed Neural Network (PINN). The neural network is trained to approximate the complex-valued function $E_z(x, y)$ such that the residual of this equation is minimized across the computational domain.

4 Neural Network Architecture

The Physics-Informed Neural Network is constructed as a fully connected feed-forward neural network that approximates the solution of the scalar Helmholtz equation:

$$\nabla^2 E_z(x, y) + \omega^2 \epsilon(x, y) \, E_z(x, y) = -i\omega J_z(x, y) \tag{24}$$

We define the model as a mapping:

$$f_\theta : R^2 \to R^2, \quad f_\theta(x, y) = [E_r(x, y), \, E_i(x, y)] \tag{25}$$

where θ denotes the network parameters, and the complex field is expressed as:

$$E_z(x, y) = E_r(x, y) + iE_i(x, y) \tag{26}$$

The network takes as input spatial coordinates (x, y), and outputs the real and imaginary components of the electric field. Each hidden layer applies a periodic activation function.

We use sinusoidal activation functions as proposed in the SIREN (Sinusoidal Representation Networks) architecture:

$$\sigma(z) = \sin(\omega_0 z) \tag{27}$$

where $\omega_0 \in R$ is a hyperparameter that controls the frequency of the activations in the first layer.

This choice is motivated by the oscillatory nature of wave equations: the solutions to the Helmholtz equation are typically superpositions of trigonometric functions, and thus:

$$E_z(x, y) \sim \sum_k A_k \sin(\mathbf{k} \cdot \mathbf{r} + \phi_k) \tag{28}$$

where $\mathbf{r} = (x, y)$, and A_k, ϕ_k are amplitude and phase terms. The sine activation allows the network to naturally represent such high-frequency components.

An important property of the sine function is that it remains stable under differentiation:

$$\frac{d}{dz} \sin(\omega_0 z) = \omega_0 \cos(\omega_0 z), \qquad \frac{d^2}{dz^2} \sin(\omega_0 z) = -\omega_0^2 \sin(\omega_0 z) \tag{29}$$

This is crucial for PINNs, where second-order derivatives like $\nabla^2 E_z$ are computed via automatic differentiation. The use of smooth periodic functions leads to numerically stable gradients during training.

The neural network used in this work consists of four hidden layers with 128 neurons each, using sinusoidal activations as in the SIREN framework. This architecture was chosen to provide sufficient expressiveness to capture the oscillatory behavior inherent in solutions to the Helmholtz equation. The number of layers and neurons was selected through a combination of theoretical considerations and empirical validation. In particular, wave-like PDEs such as the Helmholtz and Schrödinger equations are known to require high-capacity models capable of resolving fine-scale features. As shown in the original SIREN paper by Sitzmann et al. [1], sinusoidal networks with 4 to 6 hidden layers achieve state-of-the-art results in learning implicit representations of complex, high-frequency signals. Our experiments confirmed that using five layers was sufficient to model the field structure accurately while maintaining stable convergence.

No explicit weight regularization (such as L_1 or L_2 penalties) was applied during training. This is a common practice in PINNs, since the network is not trained on noisy or overfitted data, but rather on satisfying a PDE that encodes the governing physics. In this case, the residual of the Helmholtz equation itself acts as a form of regularization, promoting smoothness and physical consistency in the solution. The smooth nature of the sine activation functions also inherently discourages overfitting or oscillatory artifacts.

Training was carried out in two phases. First, Adam with initial learning rate 10^{-3}, a cosine annealing schedule, gradient clipping, and early stopping (up to 20,000 iterations). Then, a limited-memory BFGS (LBFGS) refinement for several hundred iterations to further reduce the residual. Each training step of the PINN consists of a full forward and backward pass over all collocation points in the spatial domain. Specifically, the network takes as input the coordinates (x, y) and predicts the corresponding values of the real and imaginary parts of the electric field, $E_r(x, y)$ and $E_i(x, y)$, respectively. Using automatic differentiation, the second-order derivatives $\nabla^2 E_r$ and $\nabla^2 E_i$ are computed directly from the network output. These derivatives are then used to evaluate the residual of the complex-valued Helmholtz equation. The residuals at each point are aggregated into a scalar loss value using the mean squared error formulation. Finally, a single optimization step (e.g., using the Adam optimizer) is performed to update the network parameters in a direction that reduces the total residual loss. This process is repeated for a fixed number of steps, each representing a full pass over the entire training domain.

Adam is widely used in PINN literature due to its ability to adapt learning rates per parameter, which helps stabilize training when computing higher-order derivatives through automatic differentiation.

Early stopping was employed on the physics-based residual with a small threshold, stopping when further improvement was negligible. Convergence was also assessed by visualizing the fields and verifying consistent residual decay. Instead, the training objective is to minimize a physics-based residual across the domain. While it is possible to monitor the residual loss for convergence, setting a general threshold for early stopping is problem-dependent and difficult to tune robustly. A fixed number of iterations was therefore used, with the convergence assessed by visualizing the field and verifying that the residual loss dropped to acceptable levels.

During training, collocation points are resampled each epoch: $N_{\text{interior}} =$ 40,000 interior points (with a fraction drawn from a narrow Gaussian around the source at $(0.5, 0.0)$) and $N_{\text{boundary}} = 6{,}000$ boundary points on the square $[-1, 1]^2$. A uniform grid is used only for evaluation/visualization after training. The number 128 was chosen to balance numerical resolution and computational efficiency. A higher resolution (e.g., 256 points per axis) would increase accuracy but also require more memory and training time, while a lower resolution (e.g., 64) may fail to capture fine-scale oscillatory features of the solution. With 128 points along each axis, the grid spacing is approximately $\Delta x = \Delta y =$ 0.0157, which provides sufficient coverage to enforce the PDE residual accurately across the domain. Full-batch training was feasible given the low dimensionality and modest domain size. In more complex geometries or in three-dimensional settings, mini-batch training or adaptive sampling strategies could be employed to improve efficiency and convergence.

For all simulations in this work, the angular frequency was set to $\omega = 12$, corresponding to several wavelengths between the source and the boundaries.

5 Implementation

The Physics-Informed Neural Network (PINN) was implemented in `PyTorch`, using automatic differentiation to compute the derivatives required for the

Helmholtz residual. The computational domain is the square region $[-1,1] \times [-1,1]$. During training, collocation points are *resampled at each epoch*: most points are drawn uniformly in the interior and on the boundary, and a fixed fraction are sampled from a narrow Gaussian around the source to better resolve the near field.

The relative permittivity $\epsilon(x,y)$ is defined according to the simulation case: in the free-space case, $\epsilon(x,y) = 1$ everywhere, while in the dielectric-circle case, $\epsilon(x,y) = 2$ inside a circle of radius $r = 0.3$ centered at the origin and $\epsilon(x,y) = 1$ elsewhere. The impressed source current $J_z(x,y)$ is modeled as a Gaussian distribution centered at $(x_0, y_0) = (0.5, 0.0)$:

$$J_z(x,y) = \exp\left(-\frac{(x-x_0)^2 + (y-y_0)^2}{2\sigma^2}\right), \tag{30}$$

with $\sigma = 0.05$. The angular frequency is fixed at $\omega = 8$ for all simulations.

The network maps spatial coordinates (x,y) to the real and imaginary parts of the electric field (E_r, E_i), following the SIREN architecture. It consists of an input layer with two neurons, *four* hidden layers with *128* neurons each using sinusoidal activations $\sin(\omega_0 z)$, and an output layer with two neurons representing (E_r, E_i) with linear activation. The SIREN weight initialization scheme is used to preserve stability when representing high-frequency oscillations.

The loss function is the mean squared residual of the complex Helmholtz equation

$$\nabla^2 E_z + \omega^2 \epsilon(x,y)\, E_z = -i\omega J_z(x,y), \tag{31}$$

evaluated at stochastically sampled collocation points in the interior. Writing $E_z = E_r + iE_i$ and separating into real and imaginary parts yields two coupled scalar PDEs. Automatic differentiation computes the required second-order derivatives, which are used to form the Laplacian in the residuals. The total interior loss is the mean squared sum of the real and imaginary residuals.

An open boundary is enforced using the Sommerfeld radiation condition:

$$\left(\frac{\partial}{\partial n} + ik\right) E_z = 0, \tag{32}$$

applied along uniformly sampled boundary points, where $k = \omega\sqrt{\epsilon}$. Since the boundary lies in free space, $\epsilon = 1$ and $k = \omega$. The normal derivative $\partial/\partial n$ is computed via automatic differentiation, and the boundary residual is weighted and added to the total loss.

Training is performed in two phases. First, the Adam optimizer is used with an initial learning rate of 10^{-3}, a cosine annealing schedule, gradient clipping, and early stopping, for up to 20,000 iterations. Then, a limited-memory BFGS (LBFGS) optimization is applied as a refinement step (several hundred iterations) to reduce the residual further. After training, the network is evaluated on a uniform grid for visualization; we plot the real and imaginary parts to analyze the field distribution.

6 Results and Discussion

We present PINN solutions to the 2D Helmholtz problem in two scenarios: (i) a Gaussian source in free space, and (ii) the same source in the presence of a dielectric circle of relative permittivity $\epsilon = 2$ (radius $r = 0.3$) centered at the origin. In all runs we used $\omega = 12$ and a Gaussian source width $\sigma = 0.05$, with the Sommerfeld radiation condition enforced on the square boundary.

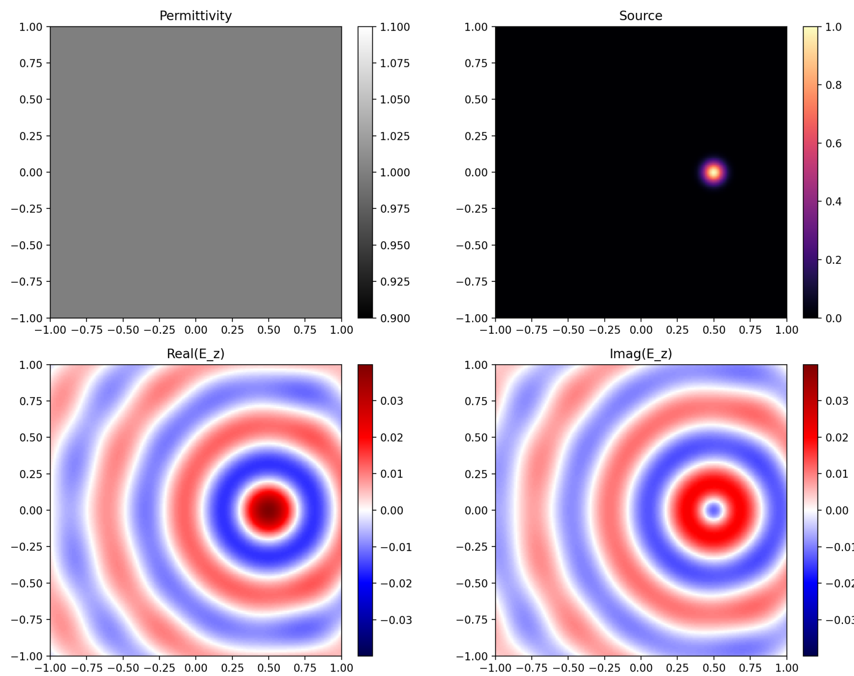

Figure 1: **Free-space case** ($\epsilon \equiv 1$, $\omega = 12$). The figure contains four panels generated by the code: **(a)** Permittivity $\epsilon(x, y)$ (uniform, shown as a flat gray field). **(b)** Source $J_z(x, y)$ (Gaussian, $\sigma = 0.05$) centered at $(0.5, 0.0)$. **(c)** Real part $\Re\{E_z(x, y)\}$. **(d)** Imaginary part $\Im\{E_z(x, y)\}$. Panels (c,d) use a zero-centered diverging colormap (blue–white–red): *red* denotes positive values, *blue* negative values, and *white* is near zero. The solution shows circular wavefronts centered at $(0.5, 0.0)$ radiating outward, with $\Re\{E_z\}$ and $\Im\{E_z\}$ approximately in spatial quadrature (phase shift $\approx \pi/2$). Amplitude decays smoothly with radius, and no spurious edge reflections are visible, indicating the radiation boundary is effective.

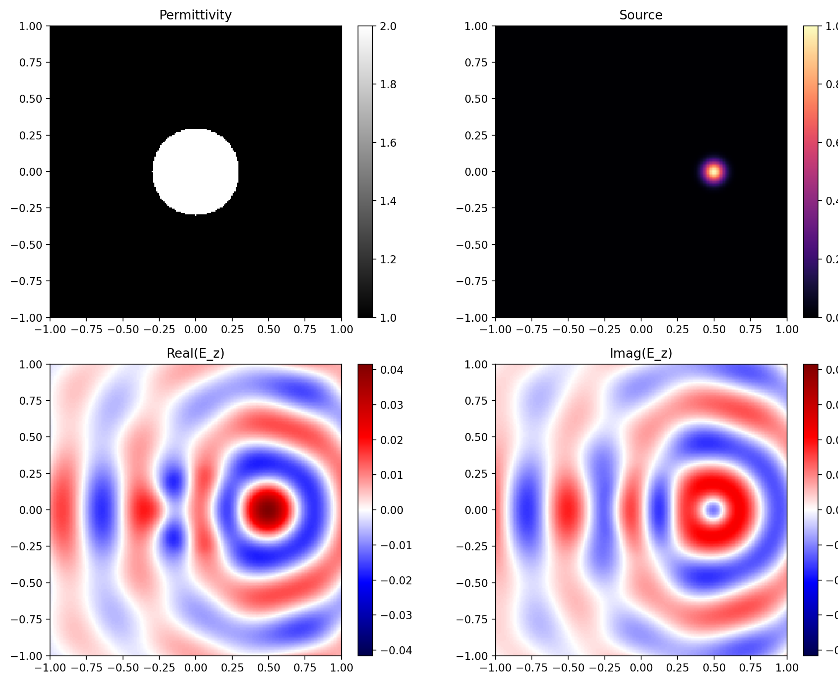

Figure 2: **Dielectric-circle case** ($\epsilon = 2$ inside $r = 0.3$, $\epsilon = 1$ elsewhere). Four panels: **(a)** Permittivity map highlighting the circular inclusion (white disk inside gray background). **(b)** Source $J_z(x, y)$ (same Gaussian as in Fig. 1). **(c)** $\Re\{E_z\}$. **(d)** $\Im\{E_z\}$. As waves enter the higher-permittivity region, the local wavelength shortens and phase is shifted, producing a slight compression of the fringes within the circle and mild scattering at the interface. The diverging colormap (blue–white–red) again indicates sign: red > 0, blue < 0. The qualitative features—refraction at the interface and weak reflected ripples—are consistent with expectations from frequency-domain electromagnetics.

The computational domain is defined as the square region $[-1, 1] \times [-1, 1]$ in normalized units, with x and y denoting spatial coordinates in the simulation plane. The source is placed at $(0.5, 0.0)$, and all distances are expressed in this normalized coordinate system. The choice of $[-1, 1]$ is purely a modeling convenience and corresponds to a finite physical size after normalization, enabling the imposition of boundary conditions at a fixed distance from the source.

The color scale in the plots represents the real part of the complex electric field $E_z(x, y)$. Since E_z is a complex quantity, its real part can take both positive and negative values. Positive values (typically shown in red) indicate that the field is in phase with the source, while negative values (shown in blue) indicate a phase shift of approximately π (out-of-phase). This sign change is a direct consequence of the oscillatory nature of solutions to the Helmholtz equation and is not an error. If the magnitude $|E_z|$ were plotted instead, the values would be strictly non-negative, but phase information would be lost.

In free space (Fig. 1), the field behaves like an outgoing 2D cylindrical wave excited by the localized source: circular symmetry, evenly spaced phase fronts, and $\Re\{E_z\}/\Im\{E_z\}$ in quadrature. Introducing the dielectric circle (Fig. 2) perturbs this symmetry: inside the inclusion, the increased ϵ reduces phase velocity so phase fronts appear slightly closer together; at the boundary, partial reflection and refraction generate subtle interference fringes. These patterns are visible in both the real and imaginary components and align with the governing physics.

Throughout, the colorbars in (c,d) are symmetric about zero, so equal-intensity red/blue correspond to equal-magnitude positive/negative field values. This makes sign and phase changes visually comparable across the two cases.

References

[1] Vincent Sitzmann, Julien N. P. Martel, Alexander W. Bergman, David B. Lindell, and Gordon Wetzstein. Implicit neural representations with periodic activation functions. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.